

PACT for Entire

Evidence-backed regression review

When the dependency graph is incomplete

Confirmed intent. Executed checks. Replayable evidence.

The unchanged caller still breaks

A new public-preview feature changes a shared helper.

An unchanged export entrypoint inherits the regression.

Changed-file selection misses both guest-export failures.

PACT ties approved policy to structural evidence and checks actual behavior.

Measured detection, locally and on Databricks

D0 → D1 strategy	Cases executed	Failures found
Changed file	2	0
Graph + fallback	10	2
All registered	10	2
Full matrix	24	2
Corrected D2	10	0 (10 pass)

24-case synthetic fixture. Same confirmed policies. Exact assertion parity.

The benefit demonstrated is detection, not faster execution or complete coverage.

Curveball: qualify uncertainty, then verify

Dynamic dispatch can omit the relevant call edge.

- 1 Keep confirmed structural paths and label heuristic evidence.
- 2 Mark incomplete analysis and run all registered approved checks.
- 3 Preserve that context in the UI, replay and Delta receipts.

D2 can pass while analysis remains partial. No invented call path.

A working review and correction flow

The screenshot displays the P/CT FOR ENTIRE Workbench interface. On the left is a dark sidebar with the logo and navigation links: 'Change review', 'Intent registry', 'Checkpoint sources', and 'Run history'. The main area shows a summary of 4 requirements, 10 candidate scenarios, 8 passing assertions, and a total time of 1.22s. Below this, two specific requirement entries are shown, both marked as 'confirmed regression'. The first entry, 'R1@2 · guest-export-own-public', details behavioral evidence and a structural path violation. The second entry, 'R1@2 · guest-export-other-public', follows a similar pattern. A third section titled 'Graph is evidence' is partially visible, indicating a 'Potentially partial analysis'.

P/CT FOR ENTIRE

WORKBENCH

- Change review
- Intent registry
- Checkpoint sources
- Run history

● E2 · Graph Intelligence
BTW Buildathon 2026
Seeded, reproducible plot

4
Requirements selected

10
Candidate scenarios executed / 24

8 → 8
Passing assertions · base → head

1.22s
Observed total · cache disabled

R1@2 · guest-export-own-public confirmed regression

Behavioral evidence · no Graph relationship asserted

No resolved structural path. This violation was demonstrated by executed checks.

Baseline: **pass** Candidate: **fail** Expected deny · observed allow

► Inspect evidence, source references & exact path

R1@2 · guest-export-other-public confirmed regression

Behavioral evidence · no Graph relationship asserted

No resolved structural path. This violation was demonstrated by executed checks.

Baseline: **pass** Candidate: **fail** Expected deny · observed allow

► Inspect evidence, source references & exact path

Graph is evidence Potentially partial analysis

3 confirmed structural paths · 0 heuristic paths. Static relationships do not prove runtime reachability.

Conservative fallback executed · All confirmed registered checks were selected. 2 additional requirements were included without inventing Graph paths.

Inspect intent

Run D1

Explain failures

Review D2

Replay evidence

Actual local workbench capture; no generated UI.

Verified implementation, inspectable evidence

22 passing tests · **5 successful cloud jobs**

Clean installed replay; receipt integrity; idempotence; Delta recovery.

Authentic final-review Checkpoint: 90d4a736f01e

Scope: registered-policy synthetic fixture. Historical policy/pre-noon capture gaps remain disclosed. Workspace jobs require permission; public reports and offline replay are available.

github.com/Shaurya002800/entire-graph/tree/pact/implementation